Atelia Feasibility Review

Assessment of requested tech pack improvements, based on client feedback and internal review

This document reviews each requested tech pack improvement, describes what implementing it would involve, and rates its complexity, based on how the current system is built today.

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1. Automatic Fabric Type Detection

Client's Request

The brief asks for the fabric type to be detected automatically from the uploaded garment images. If the user doesn't know the fabric, the AI should identify it from a list such as Jersey, Rib, French Terry, Interlock, Oxford, Poplin, Twill, Denim, and similar types, and automatically generate the Fabric Type, Fabric Construction (Knit/Woven), Suggested GSM, and Suggested Composition. Instead of asking technical questions, the brief suggests a simple question: "What best describes your fabric?" with options such as Soft T-shirt fabric, Hoodie/Sweatshirt fabric, Ribbed fabric, Shirt fabric, Denim, Sports fabric, or "I don't know" (recommended), with an optional close-up fabric photo upload.

This request is also connected to the Bosforus feedback: the tech pack stated Twill (a woven fabric) when the design was meant to be a knit fabric such as Rib, and the ask is for the spec sheet to auto-generate the correct fabric type and GSM based on the actual design.

Scope

Replacing the current fabric text field with a short set of plain-language choices — Soft T-shirt, Hoodie, Ribbed, Shirt, Denim, Sports, or "I don't know" — is achievable, but it still means reworking the existing fabric input screen, the controller logic behind it, and the fabric-default mapping table the rest of the tech pack generation already depends on, none of which exists today as an isolated piece that can be swapped out on its own. Rolling this out also requires re-testing the fabric-to-composition and fabric-to-GSM mapping across every garment type already supported, to confirm nothing currently working is disrupted.

Complexity:
6.0 / 10.0

Issue

Reading the fabric type automatically from an uploaded photo is a separate, larger problem on top of this. A photograph only shows surface texture and color — it cannot show fabric weight, stretch, the knit or weave structure, or the exact composition percentages, which are the things that actually determine what a fabric is. If the AI does happen to name the correct fabric, it would be closer to a lucky guess than something calculated with real confidence, and every attempt also uses additional AI processing at extra, recurring cost. This also connects to the earlier feedback about the tech pack stating Twill when the design was meant to be a knit fabric like Rib — that mismatch would only partially be resolved, since the same limitation would still apply going forward.

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2. Accurate Logo Placement

Client's Request

Instead of displaying a general position such as "Left Chest," the tech pack should display precise measurements on the CAD drawing — for example, "90 mm below shoulder seam" or "85 mm from center front." The brief proposes asking for Logo Placement (Left Chest, Right Chest, Center Chest, Sleeve, Back Neck, or Custom), then asking whether to use standard placement (Yes, recommended) or customize it. If customized, the user provides the distance from the neckline in millimeters and the distance from center front in millimeters, and the AI automatically annotates these dimensions on the CAD drawing. Bosforus separately asked for the same precision — e.g., "4 inches down from neck opening" — instead of a general position.

Scope

The underlying idea here — an exact distance from a reference point instead of a general area — does not depend on the AI visually interpreting anything, which makes it achievable, but delivering it still means reworking the logo placement input, the way that value is passed into the drawing-generation process, and the drawing/annotation logic itself, so it can place and label a dimension accurately rather than follow a vague text instruction. The standard-versus-custom flow described in the brief would also need to be built and tested across every logo position and garment type currently supported before it could be relied on in production.

Complexity:
7.0 / 10.0

Issue

We have not identified a technical blocking issue with this item in the way we have with the AI-detection items on this list, since placement depends on a number rather than something the AI has to visually interpret. That said, this still requires full implementation and testing work across the drawing pipeline before it can be considered dependable, and should not be assumed to be a small change.

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3. Embroidery Thread Colors

Client's Request

The factory needs the embroidery thread color. The brief proposes asking Embroidery Thread: "Same as uploaded artwork" (recommended) or "Custom thread color." If custom, the user chooses a color standard (COATS, Pantone, or Custom Color) and enters the corresponding code, and the generated tech pack includes the selected thread color reference. Bosforus separately noted there is currently no way to specify thread color at all, and suggested either the buyer selects a color, or the system references a standard color code system such as COATS (a COATS reference PDF is available to integrate), or the customer is prompted to specify a color if no code is given.

Scope

Supporting a manually entered COATS, Pantone, or custom code is achievable, but it still requires new input fields, a new data model to store and carry that thread reference through the system, and changes to how the tech pack output is assembled so the reference is displayed correctly and consistently across every generated document.

Complexity:
5.5 / 10.0

Issue

The "Same as uploaded artwork" option carries the same reliability problem as automatic fabric detection above — the AI would need to read an exact color from an image, and color can shift depending on lighting, image compression, and screen or print differences, so a color read this way would only ever be a starting suggestion rather than a dependable match. Delivering this option to a usable standard would require additional validation work on top of the manual-entry path.

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4. Automatic Care Label Generation

Client's Request

Factories require correct care instructions depending on the fabric. The brief proposes asking who should decide the care label: "Generate automatically" (recommended), "Factory recommendation," or "I'll provide my own," with the AI generating washing symbols and care instructions based on the detected fabric composition. Bosforus separately noted that care label logic is currently missing or generic, and suggested either building fabric-type-based care label logic automatically, or asking factories to suggest the correct care label per fabrication.

Scope

Care label generation is achievable, but it still requires building a new rules table that maps fabric compositions to the correct washing symbols and care instructions, integrating that table into the tech pack generation process, and validating the output against a wide range of composition combinations before it can be relied on for a factory-facing document. All three options in the brief — generate automatically, use a factory recommendation, or let the user provide their own — would each need their own input handling and validation built out.

Complexity:
4.5 / 10.0

Issue

We have not identified a technical blocking issue with this item, since it depends on fabric composition data rather than the AI interpreting an image. Even so, this represents new logic and a new dataset that does not exist in the system today, and it would need to be built, populated, and verified before it could be relied on.

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5. Better Measurement Generation (POM Table)

Client's Request

Rather than asking users to manually enter technical measurements, the brief proposes collecting simple information instead: Fit (Slim, Regular, Relaxed, Oversized, Boxy), Gender (Men, Women, Unisex, Kids), Reference Size, Model Height, and Model Weight, with an optional upload of a garment the user already owns and likes. From this, the AI should automatically generate a POM Table, measurement specifications, tolerances, and size grading.

Scope

The measurements the system already produces today are not random values — they already come directly from the AI's own evaluation of the garment — and the inputs described (Fit, Gender, Reference Size, Model Height, Model Weight) can be collected and fed into that process, which makes this achievable in principle. Building this out would still mean introducing new input handling, a new grading data structure, and a new generation path alongside the one already in place.

Complexity:
8.0 / 10.0

Issue

What remains unresolved is how a full graded table, with tolerances and a distinct value for every size, would actually be calculated from those inputs. They describe the person and the general fit, but they do not by themselves explain how the specific numeric values at every size would be derived, or how that process would relate to what the AI already generates today. This ambiguity means the scope and effort of this item cannot be pinned down with real accuracy, and it should be assumed to carry substantial additional design work beyond the other items on this list.

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6. AI Validation Before PDF Generation

Client's Request

Before generating the tech pack, the brief asks the AI to automatically check that the fabric matches the garment type, the fabric construction is correct, the GSM is realistic, the composition is coherent, the stitching matches the fabric, the logo application matches the fabric, the care label matches the composition, and the measurements match the selected fit. If inconsistencies are detected, the tech pack should be regenerated or the user warned before the final PDF is created.

Scope

Adding a validation step of this kind is achievable. It requires building a new checking step that sends the finished tech pack back to the AI, defining what counts as an inconsistency across fabric, construction, GSM, composition, stitching, logo application, care label, and measurements, and building the regeneration-or-warning flow that follows from it — all of which is new logic layered on top of the existing generation process.

Complexity:
7.0 / 10.0

Issue

This entire step depends on the AI's own judgment of what counts as "inconsistent," and that judgment is not always exact, so there is a real risk of the AI perceiving a slight inconsistency repeatedly and becoming stuck in a check-and-regenerate cycle. Every tech pack would also need to pass through this extra AI check, and potentially be regenerated more than once, adding a recurring processing cost on top of what it already costs to generate a tech pack today — on every single generation going forward, not as a one-time cost.

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7. Measurement Chart and 2D CAD Flat (Replacing the 3D Render)

Client's Request

Bosforus reported a mismatch between the image/sketch and the size table, and recommended replacing the 3D render with a 2D CAD sketch that shows exact measurements directly on the flat, which is much clearer for factories than a 3D visual.

Scope

Fixing the mismatch between the drawing and the measurement chart, and removing the 3D-style image in favor of a 2D flat sketch, is achievable, but it still requires reworking the drawing-generation prompt and logic, re-validating that the numbers shown on the drawing match the measurement chart across every garment type, and re-testing the new 2D output across the full range of garments currently supported.

Complexity:
7.0 / 10.0

Issue

This should be treated only as a correction to the existing drawing process, not as full factory-level precision across every future drawing — that level of precision is closer to the Ralph Lauren-style benchmark discussed below, which is a separate and considerably larger undertaking.

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8. Full Ralph Lauren-Style Tech Pack Format

Client's Request

Martin's email attached a Ralph Lauren tech pack as a reference, expressing that it would be great to build something closer to this standard, describing these tech packs as well designed and informative, and suggesting it as a benchmark for how the tech pack should be structured going forward. This reference document includes a cover page, a full bill of materials with suppliers for each component, several detailed construction pages, graded measurement tables across many sizes, and sample approval records.

Scope

Building toward this remains a long-term direction rather than a scoped, near-term item. A reference tech pack at this level includes a full bill of materials with supplier information for every component, multiple dedicated construction pages with exact measurements, and graded specification tables across many sizes — reaching that would mean introducing several entirely new data structures and document sections that do not exist in the system today.

Complexity:
9.5 / 10.0

Issue

This carries the most significant issue on this entire list. The tech pack currently doesn't collect or contain anywhere near the amount of information about a garment's technical construction that a document like this requires, so there isn't enough underlying detail today to fill one out. Pushing the AI to generate this much structured technical detail, when the underlying information it works from is far thinner than a document like this expects, would push it to its limits, and would likely produce inconsistent results even in the already-working parts of the tech pack that currently function well.

Part 2 — Updated Requirements Based on Client Feedback

Revised scope, complexity, and decisions following the client's response to the feasibility review

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1. Automatic Fabric Type Detection

Client Feedback

This request is not considered a priority at this stage. The current approach is to remain in place: users stay free to choose whatever fabric or material they want, and if the factory believes a different fabric would suit the project better, that can be discussed directly with the user and recommended at that point.

Decision

Keep as current. No engineering work is planned for this item.

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2. Accurate Logo Placement

Client Feedback

This remains a high-priority item. The goal is to clearly define the logo shape, its dimensions, and its exact placement on the garment. To simplify delivery, the proposed approach keeps the logo's visual placement on the drawing exactly as it works today, and additionally displays its dimensions and precise placement in written form within the tech pack.

Revised Scope

This narrows the original scope. Since the drawing itself no longer needs to carry an exact annotated measurement, the work shifts to capturing the placement input — whether a standard position or a custom distance from the neckline and center front — and presenting those values as a clear written measurement block alongside the existing drawing, rather than overlaying them onto the image. This still requires new input handling, storing the numeric values, and adding a formatted section to the tech pack output.

Complexity:
4.0 / 10.0

Issue

Because the measurements are written rather than pointed to directly on the drawing, there is some remaining risk that a factory has to cross-reference the text against the visual to confirm exactly which point the distances are measured from. This is a smaller risk than attempting to annotate an exact coordinate onto an AI-generated image, but it is not entirely eliminated.

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3. Embroidery Thread Colors

Client Feedback

The reliability concerns raised about automatic color detection reinforce a broader concern about extracting colors directly from the AI-generated model, since this limits the user's creative freedom and does not produce a sufficiently reliable result. The proposed replacement is a single "Pantone Reference" field, letting the user specify the exact embroidery color directly.

Revised Scope

This is now a single manual input field capturing a Pantone reference from the user, stored and displayed on the generated tech pack. The earlier multi-standard selector (COATS, Pantone, or Custom Color) and the "same as uploaded artwork" auto-detect option are both dropped, and no AI reading of the artwork is involved at any point.

Complexity:
2.0 / 10.0

Issue

No issue has been identified with this item. Because the value comes directly from the user rather than being read from an image, the reliability concerns that applied to the earlier version of this feature no longer apply.

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4. Automatic Care Label Generation

Client Feedback

The party best positioned to provide care instructions is the factory, since it is responsible for sourcing and supplying the fabric. Responsibility for care label recommendations should sit with the factory rather than with automatic generation.

Decision

Keep as current. No engineering work is planned for this item.

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5. Improved Measurement Generation (POM Table)

Client Feedback

This is considered interesting and relevant for the long-term evolution of the project, but it will not be a priority at this stage. Effort should instead focus on improvements that are more essential and quicker to deliver.

Decision

Keep as current. No engineering work is planned for this item at this stage.

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6. AI Validation Before PDF Generation

Client Feedback

The goal should be to minimize inconsistencies from the outset through highly structured prompts and effective guidance for users through the process, rather than catching them after the fact. Achieving zero inconsistencies is treated as an ideal rather than a realistic operational objective at this stage, and repeated regenerate-and-validate cycles are not affordable given the additional cost involved.

Revised Scope

This replaces the original post-generation validation step with upfront prevention work: restructuring the prompts sent to the AI so they carry clearer, more explicit constraints across fabric, construction, GSM, composition, stitching, logo application, care label, and measurements, and refining the guided question flow so users are steered toward internally consistent choices before generation happens at all. This is prompt-engineering and question-flow refinement work rather than a new AI-calling feature or regeneration loop.

Complexity:
5.0 / 10.0

Issue

This approach does not guarantee zero inconsistencies, only reduces their likelihood, so some risk of mismatched output remains. It does, however, avoid the recurring processing cost and the check-and-regenerate loop risk that the original validation approach carried.

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7. Measurement Chart and 2D CAD Flat (Replacing the 3D Render)

Client Feedback

The 3D render is considered essential to the user experience and to how factories understand the user's vision and design intent, and it should not be removed. The factory's recommendation to replace it entirely with a 2D CAD flat sketch is not being taken up at this time. What does need to be addressed is that the two visuals in the tech pack sometimes display different measurements, which leaves the manufacturer unable to tell which measurement set is the correct reference — this inconsistency must be resolved.

Revised Scope

This narrows the scope from replacing the 3D render down to fixing consistency between the two existing visuals. Both the 3D-style render and the accompanying technical/measurement content need to be generated from the same underlying measurement values, so the numbers shown never diverge, rather than reworking either visual's style.

Complexity:

5.0 / 10.0

Issue

The concerns raised earlier about removing the 3D render no longer apply, since it is being kept. This item still requires testing across garment types to confirm the two visuals stay in sync every time a tech pack is generated, rather than only in isolated cases.